



Bangladesh University of Business and Technology

Course Title: Computer Networks Lab

Course Code: CSE 320

LAB REPORT

Experiment Name: Network Cabling | Straight Through and Crossover Cable Connection using RJ45 Connector

Experiment No.: 01

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Introduction

Twisted pair cable is a common type of network wiring where two insulated copper wires are twisted together. This twisting helps reduce electromagnetic interference (EMI) and signal noise from external sources. In Ethernet networking, RJ-45 (Registered Jack-45) connectors are used to terminate twisted pair cables.

The TIA/EIA has defined two standard wiring schemes for Ethernet cables, the T568A Standard & T568B Standard. T568A Standard is coded White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown, Brown whereas T568B Standard is coded with White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.

Straight-Through Cable is used to connect different types of network devices such as a PC to a switch or hub. Both ends of the cable follow the same wiring standard, usually T568B.

Crossover Cable gets used to connect similar devices such as PC-to-PC or switch-to-switch. One end follows the T568A standard and the other end follows the T568B standard.

Tools

- UTP Cable (Cat5 or Cat6)
- RJ45 Connectors (1 Pair)
- Cable Stripper
- Scissors
- Crimping Tool
- Lan Cable Tester



Fig 1.1: UTP Cable



Fig 1.2: RJ45 Connectors



Fig 1.3: Crimping Tool



Fig 1.4: Lan Cable Tester

Color Code Sequence

Pin	Straight-Through (T568B)	Crossover (A: T568A)	Crossover (B: T568B)
1	Orange/white	White/Green	White/Orange
2	Orange	Green	Orange
3	Green/White	White/Orange	White/Green
4	Blue	Blue	Blue
5	White/Blue	White/Blue	White/Blue
6	Green	Orange	Green
7	White/Brown	White/Brown	White/Brown
8	Brown	Brown	Brown

Key observations from the sequence:

- **Straight-Through (T568B):** Both ends of the cable follow the T568B sequence. This means Pin 1 connects to Pin 1, Pin 2 to Pin 2, and so on.
- **Crossover:** The crossover cable is created by connecting a T568A end to a T568B end. The critical cross-connections are:
 - TX Pair (Pins 1 & 2) on T568B connect to the RX Pair (Pins 3 & 6) on T568A.
 - RX Pair (Pins 3 & 6) on T568B connect to the TX Pair (Pins 1 & 2) on T568A.
 - The Blue (4, 5) and Brown (7, 8) pairs remain straight-through, as they are typically not used for 10/100BASE-T data transmission.

Procedure

The following steps were meticulously followed to terminate the UTP cable and create both Straight-Through and Crossover cables:

Cable Preparation:

- A suitable length of UTP cable was measured and cut.
- Using the cable stripper, approximately 2.5 cm of the outer jacket was carefully removed from both ends of the cable, ensuring the inner twisted pairs remained undamaged.

Wire Arrangement:

- The four twisted pairs were untwisted and straightened.
- For the Straight-Through cable, the wires on both ends were arranged according to the T568B sequence.
- For the Crossover cable, one end was arranged according to the T568A sequence, and the other end was arranged according to the T568B sequence, as per the table.
- The arranged wires were trimmed uniformly to a length that would reach the front of the RJ45 connector, ensuring the cable jacket would be secured by the connector's strain relief.

Termination (Crimping):

- The wires were carefully inserted into the RJ45 connector, maintaining the correct sequence and ensuring each wire reached the end of its respective channel.
- The RJ45 connector, with the cable inserted, was placed into the crimping tool.
- The crimping tool was firmly squeezed until the ratchet mechanism released, which simultaneously pushes the 8 gold pins into the wires and secures the cable jacket for strain relief.

Testing and Verification:

- Both the Straight-Through and Crossover cables were connected to the LAN cable tester.
- The tester was activated to verify the continuity of all 8 wires and confirm the correct pin-to-pin mapping.

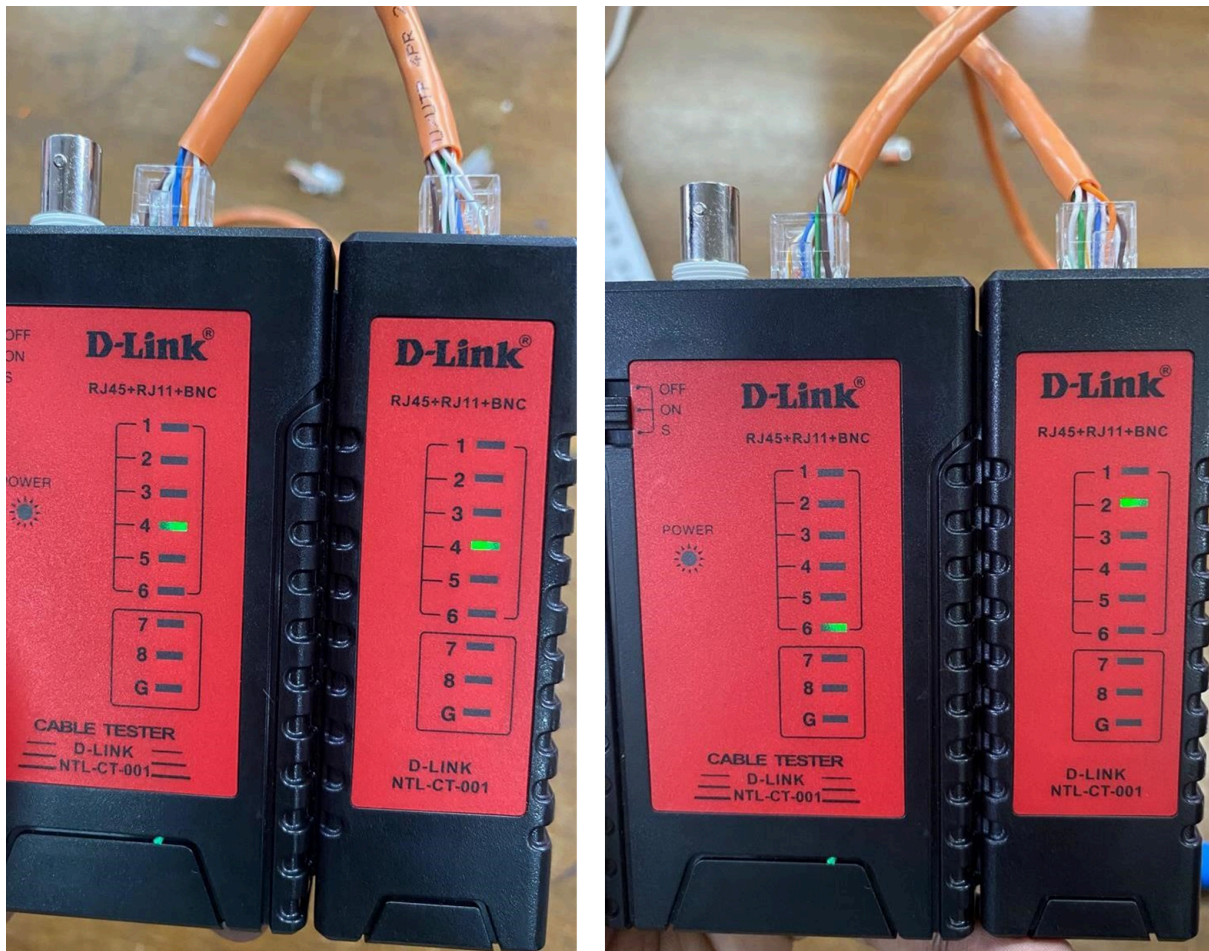


Fig 1.5: Testing the Ethernet Cable.

Conclusion

The objective of the first experiment was successfully achieved by constructing and testing both Straight-Through and Crossover Ethernet cables according to the TIA/EIA-568 standards. The experiment strengthened practical skills with networking tools and reinforced understanding of cable standards. Proper adherence to the color code sequence was critical for successful cable termination, and testing with a LAN cable tester confirmed correct functionality. This experiment provided essential foundational skills for network installation and maintenance.